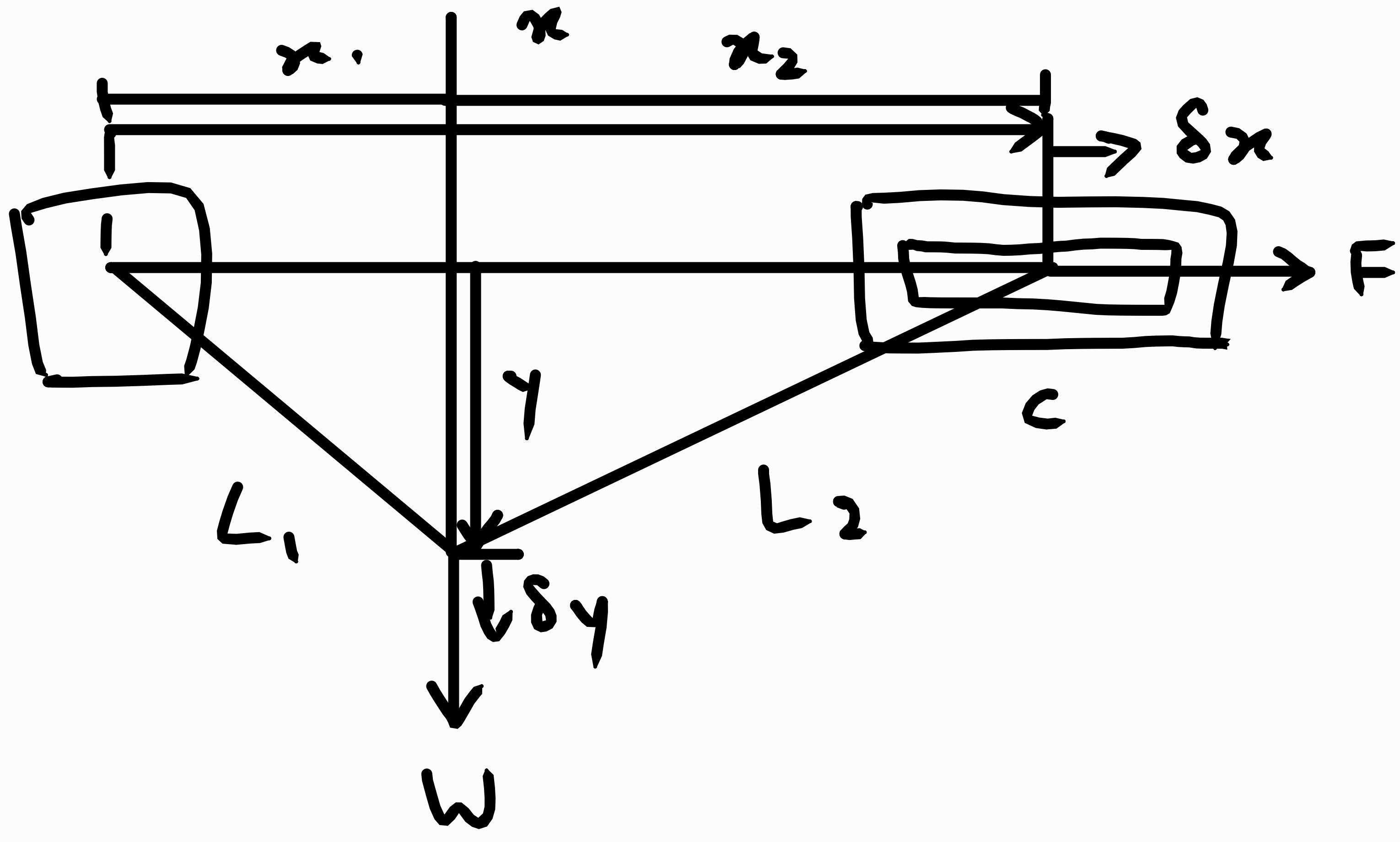


$$x = x_1 + x_2$$

$$x_1 = \sqrt{L_1^2 - y^2} \qquad x_2 = \sqrt{L_2^2 - y^2}$$

$$\therefore x = \sqrt{L_1^2 - y^2} + \sqrt{L_2^2 - y^2}$$

$$\frac{dx}{dy} = (L_1^2 - y^2)^{-\frac{1}{2}}(-2y)\left(\tfrac{1}{2}\right) + (L_2^2 - y^2)^{-\frac{1}{2}}(-2y)\left(\tfrac{1}{2}\right)$$

$$= -y(L_1^2 - y^2)^{-\frac{1}{2}} - y(L_2^2 - y^2)^{-\frac{1}{2}}$$

$$\delta x = \frac{dx}{dy}\delta y$$

$$= \left(\frac{-y}{\sqrt{L_1^2 - y^2}} + \frac{-y}{\sqrt{L_2^2 - y^2}}\right)\delta y$$

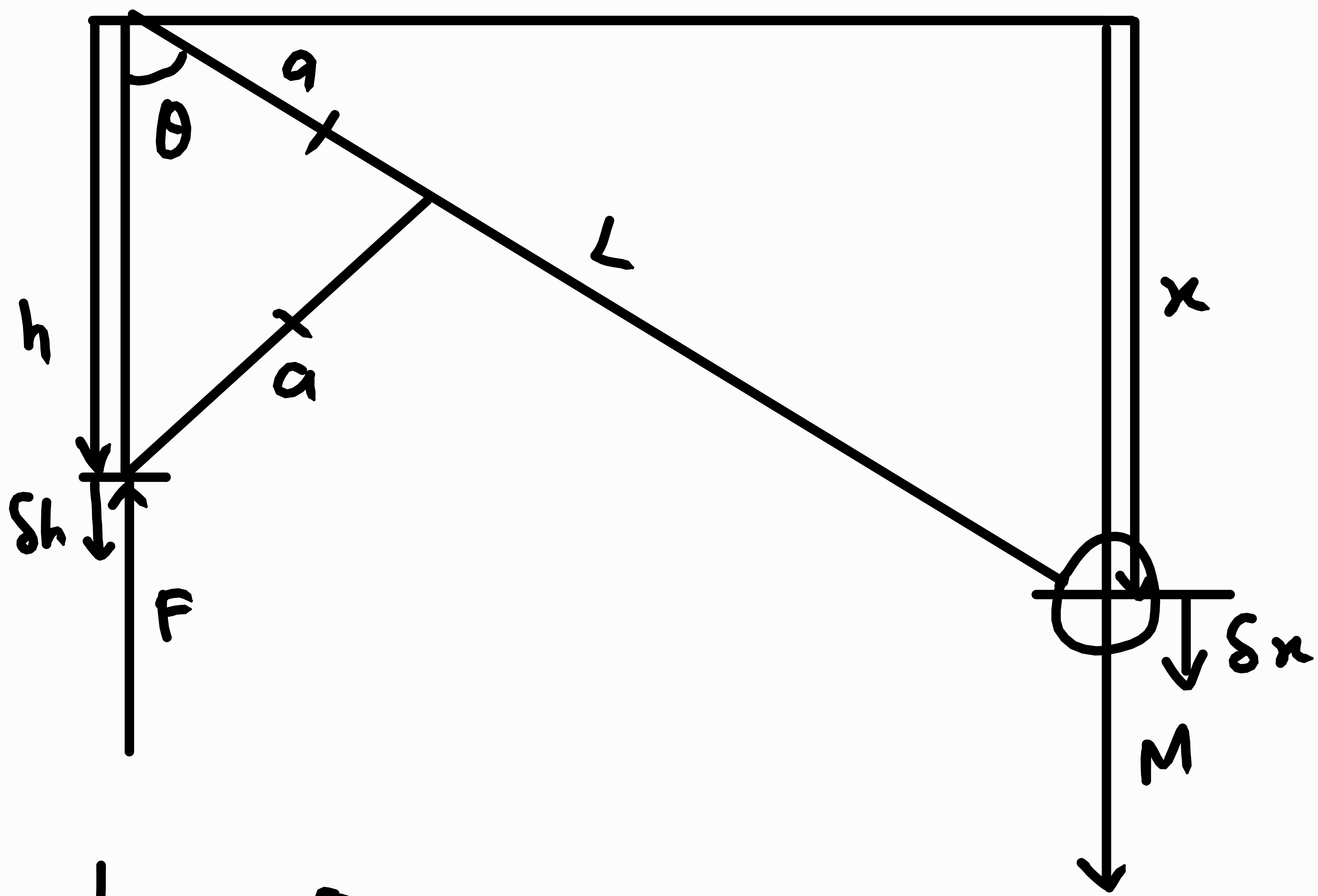
$$\delta W = 0$$

$$F\delta x + W\delta y = 0$$

$$F\left(\frac{-y}{\sqrt{L_1^2 - y^2}} + \frac{-y}{\sqrt{L_2^2 - y^2}}\right)\cancel{\delta y} + W\cancel{\delta y} = 0$$

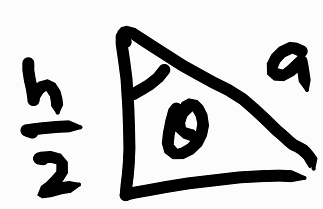
$$F = \frac{W}{y\left(\frac{1}{\sqrt{L_1^2 - y^2}} + \frac{1}{\sqrt{L_2^2 - y^2}}\right)}$$

2)



$x = L\cos\theta$

$h = 2a\cos\theta$



$\cos\theta = \frac{h/2}{a}$

$2a\cos\theta = h$

$\delta x = \frac{dx}{d\theta}\delta\theta$

$= -L\sin\theta\,\delta\theta$

$\delta h = \frac{dh}{d\theta}\delta\theta$

$= -2a\sin\theta\,\delta\theta$

$\delta W = 0$

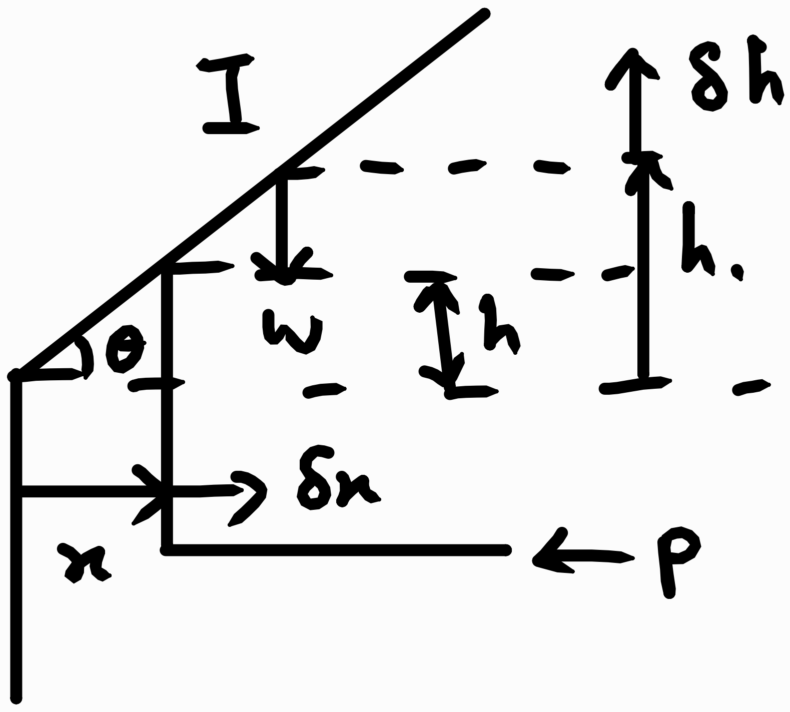
$-F\delta h + Mg\delta x = 0$

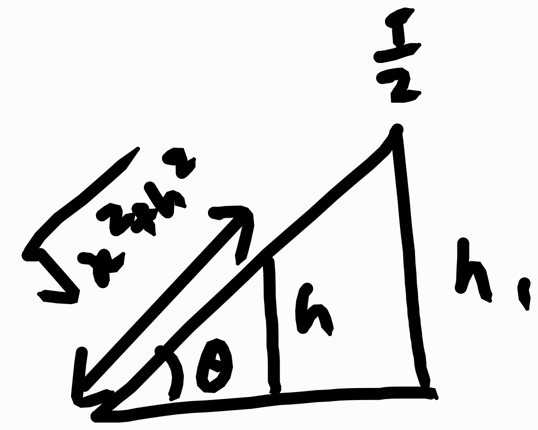
$-F(-2a\cancel{\sin\theta\,\delta\theta}) + Mg(-L\cancel{\sin\theta\,\delta\theta}) = 0$

$2aF - LMg = 0$

$F = \frac{MgL}{2a}$

3)





$$\frac{h_1}{h} = \frac{\frac{I}{2}}{\sqrt{x^2+h^2}}$$

$$h_1 = \frac{Ih}{2\sqrt{x^2+h^2}}$$

$$\delta h_1 = \frac{dh_1}{dx}\delta x$$

$$= \frac{Ih}{2}\left(x^2+h^2\right)^{-\frac{3}{2}}\left(-\frac{1}{2}\right)(2x)\,\delta x$$

$$= \frac{\cancel{2}xIh}{\cancel{2}}\left(-\frac{1}{2}\right)\left(x^2+h^2\right)^{-\frac{3}{2}}\delta x$$

$$= -\frac{xIh}{2}\left(x^2+h^2\right)^{-\frac{3}{2}}\delta x$$

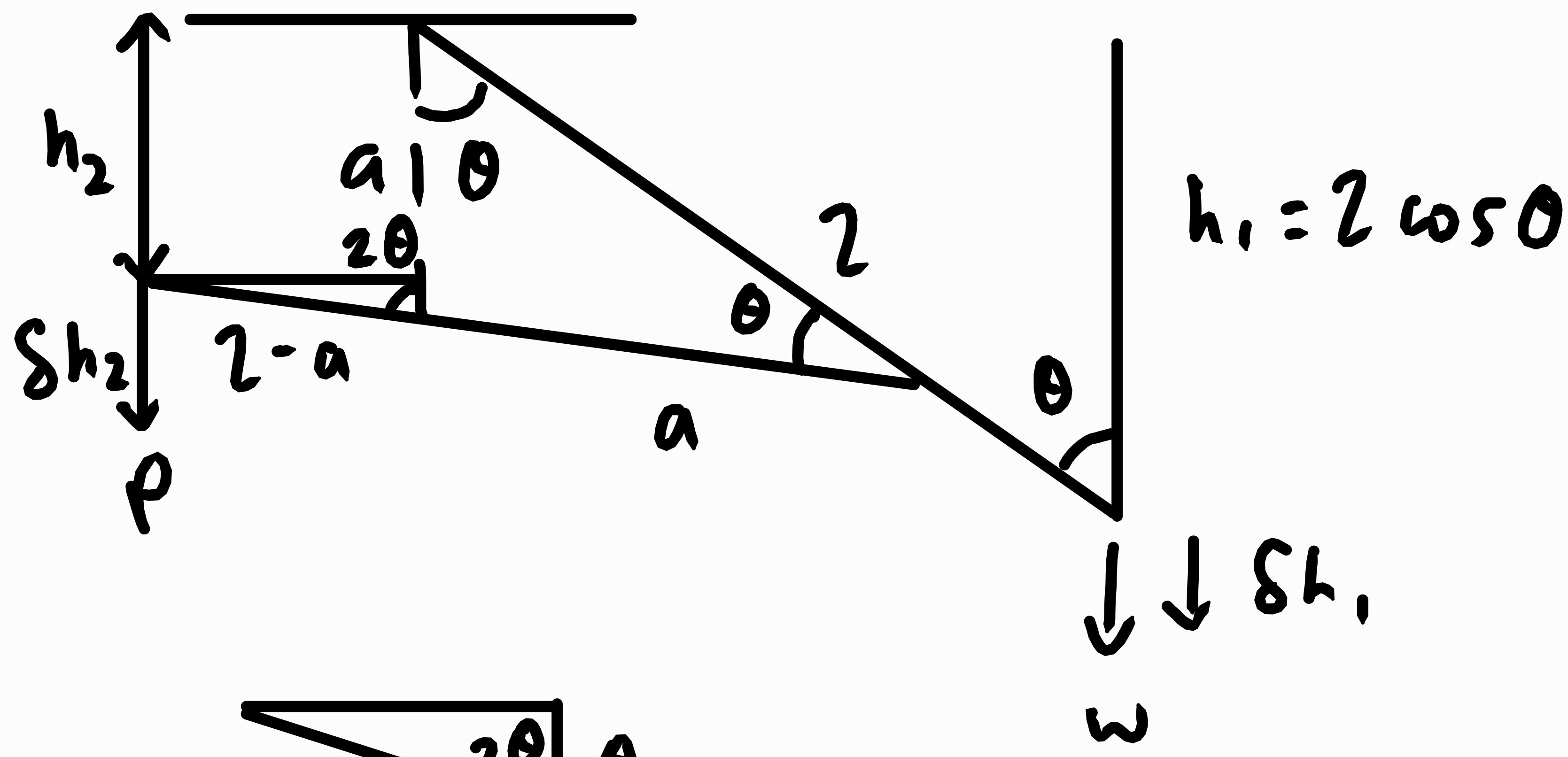
$$\delta W = 0$$

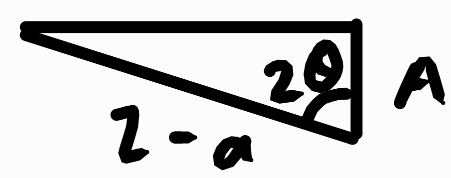
$$-W\delta h_1 - P\delta x = 0$$

$$P\cancel{\delta x} + W\left(-\frac{xIh}{2}\left(x^2+h^2\right)^{-\frac{3}{2}}\right)\cancel{\delta x} = 0$$

$$P = \frac{WxIh}{2\left(x^2+h^2\right)^{\frac{3}{2}}}$$

4)





$$\cos 2\theta = \frac{A}{2-a}$$

$$A = (2-a)\cos 2\theta$$

$$\therefore h_2 = a - (2-a)\cos 2\theta$$

$$h_1 = 2\cos\theta$$

$$\delta h_1 = \frac{dh_1}{d\theta}\delta\theta$$

$$= -2\sin\theta\,\delta\theta$$

$$\delta h_2 = \frac{dh_2}{d\theta}\delta\theta$$

$$= 2(2-a)\sin 2\theta\,\delta\theta$$

$$\delta W = 0$$

$$P\delta h_2 + W\delta h_1 = 0$$

$$2P(2-a)\sin 2\theta\,\cancel{\delta\theta} - W2\sin\theta\,\cancel{\delta\theta} = 0$$

$$2P(2-a)(2\cancel{\sin\theta}\cos\theta) - W2\cancel{\sin\theta} = 0$$

$$P = \frac{W2}{4(2-a)\cos\theta}$$